

Joint Expansion Planning of Distributed Generation and Distribution Networks

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Abstract—This paper addresses the multistage expansion planning problem of a distribution system where investments in the distribution network and in distributed generation are jointly considered. Network expansion comprises several alternatives for feeders and transformers. Analogously, the installation of distributed generation takes into account several alternatives for conventional and wind generators. Unlike what is customarily done, a set of candidate nodes for generator installation is considered. Thus, the optimal expansion plan identifies the best alternative, location, and installation time for the candidate assets. The model is driven by the minimization of the net present value of the total cost including the costs related to investment, maintenance, production, losses, and unserved energy. The costs of energy losses are modeled by a piecewise linear approximation. As another distinctive feature, radiality conditions are specifically tailored to accommodate the presence of distributed generation in order to avoid the isolation of distributed generators and the issues associated with transfer nodes. The resulting optimization problem is a mixed-integer linear program for which finite convergence to optimality is guaranteed and efficient off-the-shelf software is available. Numerical results illustrate the effective performance of the proposed approach.

Index Terms—Distributed generation, distribution system planning, multistage, network expansion, radiality.

NOMENCLATURE

A. Indices

b	Load level index.
k, κ	Indices for investment alternatives.
l	Feeder type index.
p	Generator type index.
r, s	Node indices.
t, τ	Time stage indices.
tr	Transformer type index.
ν	Index of the blocks used in the piecewise linearization for energy losses.

B. Sets

B	Set of load levels.
K^l, K^p, K^{tr}	Sets of available alternatives for feeders, generators, and transformers.
L	Set of feeder types. $L = \{EFF, ERF, NRF, NAF\}$ where EFF , ERF , NRF , and NAF denote existing fixed feeder, existing replaceable feeder, new replacement feeder, and new added feeder, respectively.
P	Set of generator types. $P = \{C, W\}$ where C and W stand for conventional and wind generation, respectively.
T	Set of time stages.
TR	Set of transformer types. $TR = \{ET, NT\}$ where ET and NT denote existing transformer and new transformer, respectively.
Υ^l	Set of branches with feeders of type l .
$\Omega_s^l, \Omega_t^{LN}, \Omega^N, \Omega^p, \Omega^{SS}$	Sets of nodes connected to node s by a feeder of type l , load nodes, system nodes, candidate nodes for distributed generation, and substation nodes.

C. Parameters

$A_{k\nu}^l, A_{k\nu}^{tr}$	Width of block ν of the piecewise linear energy losses for feeders and transformers.
$C_k^{E,p}, C_b^{SS}, C^U$	Cost coefficients of energy supplied by generators and substations, and unserved energy.
$C_k^{I,l}, C_k^{I,NT}, C_k^{I,p}, C_s^{I,SS}$	Investment cost coefficients of feeders, new transformers, generators, and substations.
$C_k^{M,l}, C_k^{M,p}, C_k^{M,tr}$	Maintenance cost coefficients of feeders, generators, and transformers.
$D_{st}, \tilde{D}_{stb}$	Actual nodal peak demand and fictitious nodal demand.
$\bar{F}_k^l$	Upper limit for actual current flows through feeders.
$\bar{G}_k^p, \hat{G}_{skb}^W$	Rated capacities of generators and maximum wind power availability.

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$\bar{G}_k^{tr}$	Upper limit for current injections of transformers.
H	Sufficiently large positive constant.
i	Annual interest rate.
IB_t	Investment budget for stage t .
ℓ_{sr}	Feeder length.
$M_{k\nu}^l, M_{k\nu}^{tr}$	Slope of block ν of the piecewise linear energy losses for feeders and transformers.
n_{DG}, n_T, n_ν	Number of candidate nodes for installation of distributed generation, number of time stages, and number of blocks of the piecewise linear energy losses.
pf	System power factor.
$RR^l, RR^{NT}, RR^p, RR^{SS}$	Capital recovery rates for investment in feeders, new transformers, generators, and substations.
$\underline{V}, \bar{V}$	Lower and upper bounds for nodal voltages.
Z_k^l, Z_k^{tr}	Unitary impedance magnitude of feeders and impedance magnitude of transformers.
Δ_b	Duration of load level b .
$\eta^l, \eta^{NT}, \eta^p, \eta^{SS}$	Lifetimes of feeders, new transformers, generators, and substation assets other than transformers.
μ_b	Loading factor of load level b .
ξ	Penetration limit for distributed generation.

D. Variables

$c_t^E, c_t^M, c_t^R, c_t^U$	Costs of production, maintenance, energy losses, and unserved energy.
c_t^I	Amortized investment cost.
c^{TPV}	Present value of the total cost.
d_{stb}^U	Nodal unserved energy.
$f_{srktb}^l, \tilde{f}_{srktb}^l$	Actual and fictitious current flows through feeders.
$g_{sktb}^p, g_{sktb}^{tr}, \tilde{g}_{sktb}^{SS}$	Current injections corresponding to generators and transformers, and fictitious current injection at substation nodes.
v_{stb}	Nodal voltage magnitude.
$x_{srkt}^l, x_{skt}^{NT}, x_{skt}^p, x_{st}^{SS}$	Binary investment variables for feeders, new transformers, generators, and substations.
$y_{srkt}^l, y_{skt}^p, y_{skt}^{tr}$	Binary utilization variables for feeders, generators, and transformers.
$\delta_{srktb\nu}^l, \delta_{sktb\nu}^{tr}$	Current in block ν of the piecewise linear energy losses for feeders and transformers.

I. INTRODUCTION

THE remarkable operational and planning benefits of distributed generation (DG) have boosted its implementation in worldwide distribution systems. In essence, DG refers to

small-scale generation units located near consumption centers. The new context where DG comes into play calls for changes in the way distribution systems are operated and planned [1].

In this paper, we propose the incorporation of DG expansion decisions in the distribution expansion planning problem. Within a centralized framework, the distribution planner is responsible for expansion decisions in order to meet the growing demand in an economic and secure fashion. The proposed distribution expansion planning problem consists in determining the timing, location, and sizing of distribution network and generation assets to be installed in an existing network so that investment and operating costs are minimized over the planning horizon. Other decisions, such as the modification of the network topology or the interconnection of isolated systems, may also be considered part of this problem. The proposed co-optimization avoids the weaknesses associated with the functional decoupling of distribution network planning and distributed generation planning that may lead to economic losses and a suboptimal use of resources. From a practical viewpoint, the proposed joint expansion planning may be used by the distribution planner to determine suitable investments in distributed generation that would require appropriate incentives for investors for their eventual installation. Note that the design of such incentive mechanisms is beyond the scope of this paper.

Despite the considerable research effort devoted to the classical distribution expansion planning problem [1]–[11] and to the decision making associated with the investment in DG facilities [12]–[16], relatively little attention has been paid to the joint expansion problem where the set of candidate assets includes both DG units and distribution equipment such as feeders, transformers, and substations.

Relevant works addressing the expansion planning of the distribution network in the presence of distributed generation are [6], [7], and [10]. However, those works disregarded expansion decisions on distributed generation. In other words, the location and timing of distributed generation were known *ex ante*, thereby constituting parameters of the optimization process. References [17]–[20] considered DG units as investment alternatives within the framework of distribution system planning. However, the emphasis was mainly placed on the impact of DG expansion on distribution investment deferral rather than on addressing the optimization problem characterizing the joint expansion planning model. From a methodological perspective, several relevant approaches are [21]–[24]. In [21], El-Khattam *et al.* proposed a mixed-integer nonlinear programming model that was fed to a commercially available software with no guarantee of global optimality. In [22], Naderi *et al.* extended the model described in [21] and proposed a genetic algorithm. Finally, another evolution-inspired heuristic, namely evolutionary particle swarm optimization, was applied in [23] and [24] to solve the mixed-integer nonlinear programming formulation of the co-optimized expansion planning.

The methods in [21]–[24] neither acknowledge global optimality nor provide a measure of the distance to the optimum. Additionally, in [21]–[24], investment decisions associated with the distribution network are exclusively related to the reinforcement of feeders and substations, thereby disregarding the construction of new distribution assets. As a consequence of this

modeling simplification, the radial topology of the network is not subject to modification and, hence, constraints imposing the radial operation of the network are not required. Moreover, the models described in [21]–[24] only consider DG comprising conventional units.

Based on the mixed-integer linear programming framework used in [4], [6], [7], and [9] for the expansion planning of the distribution network, this paper presents a new approach to simultaneously expand the distribution network and distributed generation. Unlike [21]–[24], investment decisions comprise both the reinforcement of assets and the installation of new equipment. This modeling aspect requires the incorporation of specific constraints to impose the radial operation of the resulting network. However, the use of traditional radiality constraints such as those implemented in [4], [6], [7], [9] is insufficient since radiality is no longer guaranteed when decisions on new DG and distribution assets are jointly made. Based on the findings of [10], new radiality constraints relying on fictitious demands are imposed. As another salient feature over [21]–[24], DG comprises not only conventional units but also wind power generation.

The resulting model is formulated as a mixed-integer nonlinear programming problem where nonlinearities are related to 1) quadratic energy losses in the objective function, and 2) bilinear terms involving the products of continuous and binary decision variables in the equations associated with Kirchhoff's voltage law. Both nonlinearities are recast as linear expressions by using a piecewise linear approximation for energy losses, and integer algebra results for the bilinear terms. The proposed solution approach yields a mixed-integer linear program for which effective off-the-shelf branch-and-cut software is available [25]. Note that mixed-integer linear programming (MILP) guarantees finite convergence to the optimum while providing a measure of the distance to optimality along the solution process [26].

The main contributions of this paper are as follows:

- 1) From a modeling perspective, a new formulation is presented for multistage planning of distribution systems. Unlike previously reported works, the timing, location, and sizing of not only distribution assets but also distributed generation units are modeled while explicitly imposing the radial operation of the system.
- 2) From a methodological perspective, mixed-integer linear programming is proposed to jointly expand the distribution network and distributed generation. To the best of the authors' knowledge, there is no current literature contribution referring to the application of mixed-integer linear programming to such problem.

The rest of this paper is organized as follows. Section II describes the proposed formulation. In Section III, two test systems and the corresponding results are described as a validation of the proposed model. Some relevant conclusions are drawn in Section IV. Finally, the piecewise linearization for energy losses is provided in the Appendix.

II. PROBLEM FORMULATION

The proposed model is built on the distribution network expansion planning models described in [6], [9], and [21] wherein:

- 1) a multistage planning framework is adopted, 2) a discretization of the annual load curve into several load levels is used to characterize the demand, 3) radial operation of the distribution network is explicitly imposed, 4) an approximate network model is used, 5) costs of losses are included in the objective function, and 6) several investment alternatives exist for each asset. Unlike [6], [9], [21]–[24], it is assumed that the distribution planner has the capability to make strategic investment decisions on the joint installation of new feeders and DG including both conventional and wind power generation. In addition, it is worth emphasizing that DG location is a result of the optimization process.

A. Objective Function and Cost-Related Terms

According to [9], the optimization aims at minimizing the present value of the total cost:

$$c^{TPV} = \sum_{t \in T} \frac{(1+i)^{-t}}{i} c_t^I + \sum_{t \in T} [(1+i)^{-t} (c_t^M + c_t^E + c_t^R + c_t^U)] + \frac{(1+i)^{-n_T}}{i} (c_{n_T}^M + c_{n_T}^E + c_{n_T}^R + c_{n_T}^U). \quad (1)$$

As done in [9], expression (1) includes three terms. The first term corresponds to the present worth value of the investment cost under the assumption of a perpetual or infinite planning horizon [27]. In other words, the investment cost is amortized in annual installments throughout the lifetime of the installed equipment, considering that after the equipment lifetime has expired there is a reinvestment in an identical piece of equipment. The second term is the present value of the operating costs throughout the time stages. Finally, the third term represents the present value of the operating costs incurred after the last time stage. As can be noted, such term relies on the operating costs at the last time stage and also assumes a perpetual planning horizon.

The total cost in (1) comprises amortized investment, maintenance, production, energy losses, and unserved energy costs which are formulated as follows:

$$c_t^I = \sum_{l \in \{NRF, NAF\}} RR^l \sum_{k \in K^l} \sum_{(s,r) \in \Upsilon^l} C_k^{I,l} \ell_{sr} x_{srkt}^l + RR^{SS} \sum_{s \in \Omega^{SS}} C_s^{I,SS} x_{st}^{SS} + RR^{NT} \sum_{k \in K^{NT}} \sum_{s \in \Omega^{SS}} C_k^{I,NT} x_{skt}^{NT} + \sum_{p \in P} RR^p \sum_{k \in K^p} \sum_{s \in \Omega^p} C_k^{I,p} pf \bar{G}_k^p x_{skt}^p; \quad \forall t \in T \quad (2)$$

$$c_t^M = \sum_{l \in L} \sum_{k \in K^l} \sum_{(s,r) \in \Upsilon^l} C_k^{M,l} (y_{srkt}^l + y_{rskt}^l) + \sum_{tr \in TR} \sum_{k \in K^{tr}} \sum_{s \in \Omega^{SS}} C_k^{M,tr} y_{skt}^{tr} + \sum_{p \in P} \sum_{k \in K^p} \sum_{s \in \Omega^p} C_k^{M,p} y_{skt}^p; \quad \forall t \in T \quad (3)$$

$$c_t^E = \sum_{b \in B} \Delta_b p f \left(\sum_{tr \in TR} \sum_{k \in K^{tr}} \sum_{s \in \Omega^{SS}} C_b^{SS} g_{sktb}^{tr} + \sum_{p \in P} \sum_{k \in K^p} \sum_{s \in \Omega^p} C_k^{E,p} g_{sktb}^p \right); \quad \forall t \in T \quad (4)$$

$$c_t^R = \sum_{b \in B} \Delta_b C_b^{SS} p f \left[\sum_{tr \in TR} \sum_{k \in K^{tr}} \sum_{s \in \Omega^{SS}} Z_k^{tr} (g_{sktb}^{tr})^2 + \sum_{l \in L} \sum_{k \in K^l} \sum_{(s,r) \in \Upsilon^l} Z_k^{l, \ell_{sr}} (f_{srktb}^l + f_{rsktb}^l)^2 \right]; \quad \forall t \in T \quad (5)$$

$$c_t^U = \sum_{b \in B} \sum_{s \in \Omega_t^{LN}} \Delta_b C^U p f d_{stb}^U; \quad \forall t \in T \quad (6)$$

where capital recovery rates are computed as $RR^l = (i(1+i)^{\eta^l})/((1+i)^{\eta^l} - 1)$, $\forall l \in \{NRF, NAF\}$; $RR^{NT} = (i(1+i)^{\eta^{NT}})/((1+i)^{\eta^{NT}} - 1)$; $RR^p = (i(1+i)^{\eta^p})/((1+i)^{\eta^p} - 1)$, $\forall p \in P$; and $RR^{SS} = (i(1+i)^{\eta^{SS}})/((1+i)^{\eta^{SS}} - 1)$.

In (2), the amortized investment cost at each stage is formulated as the sum of terms related to: 1) replacement and addition of feeders, 2) reinforcement of existing substations and construction of new ones, 3) installation of new transformers, and 4) installation of DG. It should be emphasized that the investment cost for substations corresponds to the cost associated with the upgrading or construction of this infrastructure excluding the cost of transformers, which is explicitly considered in the third term of (2). Expressions (3) model the maintenance costs of feeders, transformers, and generators at each stage. Note that, for each time stage, a single binary variable per conductor in the feeder connecting nodes s and r is used to model the corresponding investment decision, namely x_{srkt}^l . In contrast, two binary variables, y_{srkt}^l and y_{rskt}^l , as well as two continuous variables, f_{srktb}^l and f_{rsktb}^l , are associated with each feeder in order to model its utilization and current flow, respectively. Note that f_{srktb}^l is greater than 0 and equal to the current flow through the feeder between nodes s and r measured at node s only when the current flows from s to r , being 0 otherwise.

The production costs associated with substations and distributed generators are characterized in (4). Similarly to [9], the costs of energy losses in transformers and feeders are modeled in (5) as quadratic terms. The minimization of the costs of energy losses constitutes another distinctive feature over the model described in [6] and is motivated by the need to avoid the undesired increase in energy losses associated with an inadequate installation of distributed generation [14], [28]. Nonlinear costs of energy losses can be accurately approximated by a set of tangent lines as explained in the Appendix. This approximation yields piecewise linear functions, which, for practical purposes, are indistinguishable from the nonlinear models if enough blocks are used. Finally, expressions (6) correspond to the penalty cost of unserved energy. The selection of a sufficiently large value for the cost coefficient C^U ensures that the unserved energy becomes zero when power balance can be met with available generation and distribution resources.

B. Kirchhoff's Laws and Operational Limits

The constraints associated with the system operation are formulated as

$$\underline{V} \leq v_{stb} \leq \bar{V}; \quad \forall s \in \Omega^N, \forall t \in T, \forall b \in B \quad (7)$$

$$0 \leq f_{srktb}^l \leq y_{srkt}^l \bar{F}_k^l; \quad \forall l \in L, \forall s \in \Omega_r^l, \forall r \in \Omega^N, \forall k \in K^l, \forall t \in T, \forall b \in B \quad (8)$$

$$0 \leq g_{sktb}^{tr} \leq y_{skt}^{tr} \bar{G}_k^{tr}; \quad \forall tr \in TR, \forall s \in \Omega^{SS}, \forall k \in K^{tr}, \forall t \in T, \forall b \in B \quad (9)$$

$$0 \leq d_{stb}^U \leq \mu_b D_{st}; \quad \forall s \in \Omega_t^{LN}, \forall t \in T, \forall b \in B \quad (10)$$

$$0 \leq g_{sktb}^C \leq y_{skt}^C \bar{G}_k^C; \quad \forall s \in \Omega^C, \forall k \in K^C, \forall t \in T, \forall b \in B \quad (11)$$

$$0 \leq g_{sktb}^W \leq y_{skt}^W \min \{ \bar{G}_k^W, \hat{G}_{sktb}^W \}; \quad \forall s \in \Omega^W, \forall k \in K^W, \forall t \in T, \forall b \in B \quad (12)$$

$$\sum_{p \in P} \sum_{k \in K^p} \sum_{s \in \Omega^p} g_{sktb}^p \leq \xi \sum_{s \in \Omega_t^{LN}} \mu_b D_{st}; \quad \forall t \in T, \quad \forall b \in B \quad (13)$$

$$\sum_{l \in L} \sum_{k \in K^l} \sum_{r \in \Omega_s^l} (f_{srktb}^l - f_{rsktb}^l) = \sum_{tr \in TR} \sum_{k \in K^{tr}} g_{sktb}^{tr} + \sum_{p \in P} \sum_{k \in K^p} g_{sktb}^p - \mu_b D_{st} + d_{stb}^U; \quad \forall s \in \Omega^N, \forall t \in T, \forall b \in B \quad (14)$$

$$y_{srkt}^l [Z_k^{l, \ell_{sr}} f_{srktb}^l - (v_{stb} - v_{rtb})] = 0; \quad \forall l \in L, \quad \forall s \in \Omega_r^l, \forall r \in \Omega^N, \forall k \in K^l, \forall t \in T, \forall b \in B. \quad (15)$$

Constraints (7) set the bounds on the magnitudes of nodal voltages. Thus, the upper bounds imposed in (7) prevent voltage increase at the end of feeders due to the installation of DG. Similarly, upper and lower limits on network- and generation-related variables are modeled in (8)–(12). Note that in (8), (9) the upper limits associated with feeders, $\bar{F}_k^l$, and transformers, $\bar{G}_k^{tr}$, may be less than the corresponding ratings in order to allow for an operating reserve that might be required under contingency [1]. Analogously, the upper bound for wind power generation is the minimum value between the generator rated capacity and the available power associated with the corresponding level of wind speed (12). DG penetration is limited in (13) as a ξ fraction of the demand.

Finally, based on the linearized network model that was first proposed by Haffner *et al.* [6] and successfully applied in [7], [9], constraints (14)–(15) represent the effect of the distribution network. As described in [6], the linearized network model is an adapted version of the dc model used for the transmission network that is based on two assumptions: 1) all current injections and flows have the same power factor, and 2) the per-unit voltage drop across a branch is equal to the difference between the per-unit magnitudes of the nodal voltages at both ends of the branch. Assumption 1) allows expressing Kirchhoff's current law as a set of linear scalar equalities in terms of current magnitudes, giving rise to nodal current balance equations (14). In addition, assumption 2) allows formulating Kirchhoff's voltage law for each feeder in use as a linear expression relating the magnitudes of currents, nodal voltages, and branch impedances.

Constraints (15) extend this result to account for the utilization state of all feeders. Based on [29], nonlinear expressions (15) have the following linear equivalents:

$$\begin{aligned} -H(1 - y_{srkt}^l) &\leq Z_k^l \ell_{sr} f_{srktb}^l \\ &- (v_{stb} - v_{rtb}) \leq H(1 - y_{srkt}^l); \\ \forall l \in L, \forall s \in \Omega_r^l, \forall r \in \Omega^N, \forall k \in K^l, \forall t \in T, \forall b \in B. \end{aligned} \quad (16)$$

We recognize that the use of the simplified network model described in [6] leads to results that may be optimistic and that a complete study of the distribution network operation should consider a full ac model. This generalization would, however, render the problem essentially intractable through optimization and would have to be solved by heuristics or repeated simulations. This modeling limitation notwithstanding, the solution of the joint expansion problem based on the linearized network model is acceptable for the purposes of distribution planning [6], [7], [9] and provides the planner with a first estimate of a cost-effective expansion scheme.

C. Investment and Utilization Constraints

Investment and utilization decisions are constrained according to the following expressions:

$$\sum_{t \in T} \sum_{k \in K^l} x_{srkt}^l \leq 1; \quad \forall l \in \{NRF, NAF\}, \forall (s, r) \in \Upsilon^l \quad (17)$$

$$\sum_{t \in T} x_{st}^{SS} \leq 1; \quad \forall s \in \Omega^{SS} \quad (18)$$

$$\sum_{t \in T} \sum_{k \in K^{NT}} x_{skt}^{NT} \leq 1; \quad \forall s \in \Omega^{SS} \quad (19)$$

$$\sum_{t \in T} \sum_{k \in K^p} x_{skt}^p \leq 1; \quad \forall p \in P, \forall s \in \Omega^p \quad (20)$$

$$x_{skt}^{NT} \leq \sum_{\tau=1}^t x_{s\tau}^{SS}; \quad \forall s \in \Omega^{SS}, \forall k \in K^{NT}, \forall t \in T \quad (21)$$

$$y_{srkt}^{EFF} + y_{rskt}^{EFF} \leq 1; \quad \forall (s, r) \in \Upsilon^{EFF}, \forall k \in K^{EFF}, \forall t \in T \quad (22)$$

$$y_{srkt}^l + y_{rskt}^l \leq \sum_{\tau=1}^t x_{srk\tau}^l; \quad \forall l \in \{NRF, NAF\}, \forall (s, r) \in \Upsilon^l, \forall k \in K^l, \forall t \in T \quad (23)$$

$$y_{srkt}^{ERF} + y_{rskt}^{ERF} \leq 1 - \sum_{\tau=1}^t \sum_{\kappa \in K^{NRF}} x_{srk\tau}^{NRF}; \quad \forall (s, r) \in \Upsilon^{ERF}, \forall k \in K^{ERF}, \forall t \in T \quad (24)$$

$$y_{skt}^{NT} \leq \sum_{\tau=1}^t x_{sk\tau}^{NT}; \quad \forall s \in \Omega^{SS}, \forall k \in K^{NT}, \forall t \in T \quad (25)$$

$$y_{skt}^p \leq \sum_{\tau=1}^t x_{sk\tau}^p; \quad \forall p \in P, \forall s \in \Omega^p, \forall k \in K^p, \forall t \in T \quad (26)$$

$$\begin{aligned} &\sum_{l \in \{NRF, NAF\}} \sum_{k \in K^l} \sum_{(s,r) \in \Upsilon^l} C_k^{I,l} \ell_{sr} x_{srkt}^l \\ &+ \sum_{s \in \Omega^{SS}} C_s^{I,SS} x_{st}^{SS} + \sum_{k \in K^{NT}} \sum_{s \in \Omega^{SS}} C_k^{I,NT} x_{skt}^{NT} \\ &+ \sum_{p \in P} \sum_{k \in K^p} \sum_{s \in \Omega^p} C_k^{I,p} p f \bar{G}_k^p x_{skt}^p \leq IB_t; \quad \forall t \in T. \end{aligned} \quad (27)$$

As per (17)–(20), a maximum of one reinforcement, replacement or addition is allowed for each system component and location along the planning horizon. Constraints (21) guarantee that new transformers can only be added in substations that have been previously expanded or built. Constraints (22)–(24) model the utilization of feeders while explicitly characterizing the direction of current flows. The utilization of new transformers and distributed generators is formulated in (25) and (26), respectively. Finally, constraints (27) impose a budgetary limit for investments at each stage.

D. Radiality Constraints

The proposed model includes the following traditional radiality constraints [4], [9]:

$$\sum_{l \in L} \sum_{s \in \Omega_r^l} \sum_{k \in K^l} y_{srkt}^l = 1; \quad \forall r \in \Omega_t^{LN}, \forall t \in T \quad (28)$$

$$\sum_{l \in L} \sum_{s \in \Omega_r^l} \sum_{k \in K^l} y_{srkt}^l \leq 1; \quad \forall r \notin \Omega_t^{LN}, \forall t \in T. \quad (29)$$

Constraints (28) impose load nodes to have a single input flow while expressions (29) set a maximum of one input flow for the remaining nodes. It is worth mentioning that, in the absence of DG, this formulation does not feature the shortcomings of that described in [6] and [10] and is suitable to impose radial operation even when transfer nodes are part of the distribution network. However, when DG is considered, (28) and (29) are insufficient to prevent the existence of areas exclusively supplied by DG and thereby topologically disconnected from all substations. This difficulty is overcome by adding the following set of radiality constraints [10]:

$$\begin{aligned} &\sum_{l \in L} \sum_{k \in K^l} \sum_{r \in \Omega_s^l} (\tilde{f}_{srktb}^l - \tilde{f}_{rsktb}^l) = \tilde{g}_{stb}^{SS} - \tilde{D}_{stb}; \\ &\forall s \in \Omega^N, \forall t \in T, \forall b \in B \end{aligned} \quad (30)$$

$$0 \leq \tilde{f}_{srktb}^{EFF} \leq n_{DG}; \quad \forall s \in \Omega_r^{EFF}, \forall r \in \Omega^N, \forall k \in K^{EFF}, \forall t \in T, \forall b \in B \quad (31)$$

$$0 \leq \tilde{f}_{srktb}^{ERF} \leq n_{DG} \left(1 - \sum_{\tau=1}^t \sum_{\kappa \in K^{NRF}} x_{srk\tau}^{NRF} \right); \quad \forall (s, r) \in \Upsilon^{ERF}, \forall k \in K^{ERF}, \forall t \in T, \forall b \in B \quad (32)$$

$$0 \leq \tilde{f}_{rsktb}^{ERF} \leq n_{DG} \left(1 - \sum_{\tau=1}^t \sum_{\kappa \in K^{NRF}} x_{srk\tau}^{NRF} \right); \quad \forall (s, r) \in \Upsilon^{ERF}, \forall k \in K^{ERF}, \forall t \in T, \forall b \in B \quad (33)$$

$$0 \leq \tilde{f}_{srktb}^l \leq n_{DG} \sum_{\tau=1}^t x_{srk\tau}^l; \quad \forall l \in \{NRF, NAF\}, \quad \forall (s, r) \in \Upsilon^l, \forall k \in K^l, \forall t \in T, \forall b \in B \quad (34)$$

$$0 \leq \tilde{f}_{rsktb}^l \leq n_{DG} \sum_{\tau=1}^t x_{srk\tau}^l; \quad \forall l \in \{NRF, NAF\}, \quad \forall (s, r) \in \Upsilon^l, \forall k \in K^l, \forall t \in T, \forall b \in B \quad (35)$$

$$0 \leq \tilde{g}_{stb}^{SS} \leq n_{DG}; \quad \forall s \in \Omega^{SS}, \forall t \in T, \forall b \in B. \quad (36)$$

As described in [10], the existence of isolated generators is avoided through (30)–(36), which model a fictitious system with fictitious demands. According to [10], the fictitious demand at

load nodes that are candidate locations for DG installation is equal to 1 p.u., whereas the fictitious demand at the remaining nodes is set to 0. Mathematically

$$\tilde{D}_{stb} = \begin{cases} 1; & \forall s \in ((\Omega^C \cup \Omega^W) \cap \Omega_t^{LN}), \forall t \in T, \forall b \in B \\ 0; & \forall s \notin ((\Omega^C \cup \Omega^W) \cap \Omega_t^{LN}), \forall t \in T, \forall b \in B. \end{cases} \quad (37)$$

Such fictitious nodal demands can only be supplied by fictitious substations located at the original substation nodes, which inject fictitious energy through the actual feeders.

Constraints (30) represent the nodal fictitious current balance equations. Constraints (31)–(35) bound fictitious flows through feeders. Constraints (36) set the limits for the fictitious currents injected by the fictitious substations. It is worth mentioning that, to the best of the authors' knowledge, the combined use of (28)–(36) to avoid the issues with transfer nodes and isolated generators is novel, thereby constituting a relevant modeling contribution of this paper.

III. NUMERICAL RESULTS

This section presents and discusses results from two case studies. For didactical purposes, a 24-node system is first analyzed over a three-year planning horizon. The scalability of the proposed approach is subsequently validated with a larger case study comprising 138 nodes and a ten-year planning horizon. The currency used in the simulations is U.S. dollars, hereinafter denoted by \$.

The following data are common for both case studies:

- The planning horizon is divided into yearly stages considering a 7.1% interest rate.
- Three load levels are considered with loading factors equal to 70%, 83%, and 100% of the corresponding peak demand, and durations equal to 2000 h/year, 5760 h/year, and 1000 h/year, respectively.
- The cost for unserved energy, C^U , is \$2000/MWh, as done in [9].
- The systems are divided into three zones according to the wind data reported in Table I. For the sake of simplicity, it is assumed that wind speed data remain unchanged along the time span.
- Investment decisions for substations consist of: 1) expanding existing substations by adding a new transformer, and 2) building a new substation from scratch. It is assumed that η^{SS} is considerably larger than the lifetimes of the other distribution assets. Thus, $RR^{SS} = i$. Based on [21], the costs of the energy supplied by all substations are identical and equal to \$57.7/MWh, \$70.0/MWh, and \$85.3/MWh, for load levels 1–3, respectively.
- Based on [12], [24], [30], and [31], the economic and technical features of candidate DG units are presented in Table II, where two alternatives are considered for each technology. Maintenance costs are set so that $C_k^{M,p} = 0.05 C_k^{I,p} pf \bar{G}_k^p$. Wind generation limits, $\hat{G}_{sktb}^W$, associated with the wind speed levels given in Table I are determined using the data available in [31] related to wind generators E44 and E82, which correspond to alternatives

TABLE I
WIND SPEED DATA FOR EACH STAGE (m/s)

Zone	Load level		
	1	2	3
A	8.53	9.12	10.04
B	6.13	7.26	7.11
C	4.13	5.10	5.56

TABLE II
DATA FOR CANDIDATE DG UNITS

Alternative	$\bar{G}_k^C$	$C_k^{I,C}$	$C_k^{E,C}$	$\bar{G}_k^W$	$C_k^{I,W}$	$C_k^{E,W}$
k	(MVA)	(\$/MW)	(\$/MWh)	(MVA)	(\$/MW)	(\$/MWh)
1	1	500000	47	0.91	1850000	0
2	2	490000	45	2.05	1840000	0

1 and 2, respectively. A 20-year lifetime is taken into account for all units.

- A three-block piecewise linearization for energy losses is considered.
- The system power factor pf is set equal to 0.9.

In order to assess the impact of DG on expansion planning, both case studies have been first solved considering that the only candidate assets for expansion are feeders, substations, and transformers. Secondly, investment in DG is allowed.

Simulations have been implemented on a Dell PowerEdge R910X64 with four Intel Xeon E7520 processors at 1.866 GHz and 32 GB of RAM using CPLEX 12.6 [25] under GAMS 24.2 [32]. The stopping criterion is based on an optimality gap equal to 1%.

A. Illustrative Example

This example is based on the 20-kV test system described in [2]. As shown in Fig. 1, the system comprises 20 load nodes, represented by circles; 4 substation nodes, depicted as squares; and 33 branches, indicated by lines. Nodal peak demands and branch lengths are presented in Tables III and IV, respectively. Voltages at substation nodes are set to 1.05 p.u. on a 20-kV base, whereas upper and lower bounds for voltages at load nodes are equal to 1.05 p.u. and 0.95 p.u., respectively.

Existing substations, which are located at nodes 21 and 22, include a 7.5-MVA transformer characterized by an impedance equal to 0.25 Ω and a maintenance cost equal to \$1000. Nodes 23 and 24 represent the locations of candidate substations. Expansion costs of substations at nodes 21–24 are \$100 000, \$100 000, \$140 000, and \$180 000, respectively. Data for candidate transformers are provided in Table V, where the same two alternatives are available for each substation. The lifetime of all candidate transformers is 15 years.

Branches are categorized as existing and non-existing. Existing branches are further classified into two types: 1) those with one feeder not subject to modification, represented by solid lines in Fig. 1; and 2) those with one feeder that can undergo replacement, depicted as solid double lines. Non-existing branches where a candidate feeder can be installed are drawn as dashed lines. The capacity and unitary impedance of existing feeders are 3.94 MVA and 0.732 Ω /km, respectively. Data for candidate conductors in branches subject to replacement

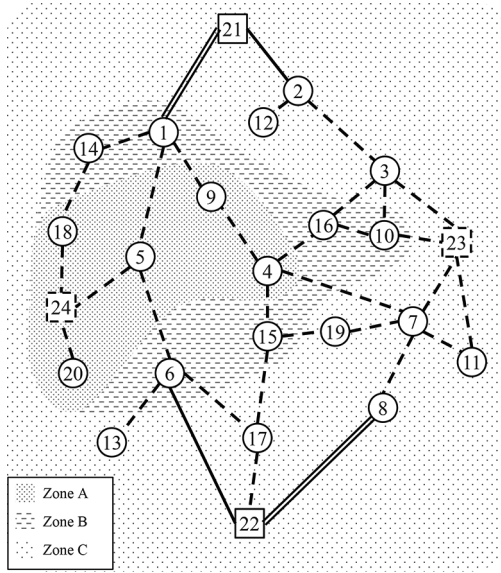


Fig. 1. One-line diagram of the 24-node system.

TABLE III
24-NODE SYSTEM—NODAL PEAK DEMANDS (MVA)

Node	Stage			Node	Stage		
	1	2	3		1	2	3
1	4.05	3.45	5.42	11	0.00	1.91	2.80
2	0.78	0.77	1.21	12	0.00	0.93	1.29
3	2.58	3.38	3.98	13	0.00	1.15	1.87
4	0.32	0.41	2.43	14	0.00	3.05	3.16
5	0.28	0.37	0.47	15	0.00	1.62	1.62
6	1.17	0.92	1.81	16	0.00	2.16	1.22
7	4.04	3.70	4.36	17	0.00	0.00	2.40
8	0.72	0.60	0.94	18	0.00	0.00	2.10
9	1.14	1.12	1.77	19	0.00	0.00	1.81
10	1.56	2.04	2.40	20	0.00	0.00	3.79

TABLE IV
24-NODE SYSTEM—BRANCH LENGTHS (km)

Branch		ℓ_{sr}	Branch		ℓ_{sr}	Branch		ℓ_{sr}
s	r		s	r		s	r	
1	5	2.22	4	9	1.20	7	23	0.90
1	9	1.20	4	15	1.60	8	22	1.90
1	14	1.20	4	16	1.30	10	16	1.60
1	21	2.20	5	6	2.40	10	23	1.30
2	3	2.00	5	24	0.70	11	23	1.60
2	12	1.10	6	13	1.20	14	18	1.00
2	21	1.70	6	17	2.20	15	17	1.20
3	10	1.10	6	22	2.70	15	19	0.80
3	16	1.20	7	8	2.00	17	22	1.50
3	23	1.20	7	11	1.10	18	24	1.50
4	7	2.60	7	19	1.20	20	24	0.90

TABLE V
24-NODE SYSTEM—DATA FOR CANDIDATE TRANSFORMERS

Alternative 1				Alternative 2			
$\bar{G}_1^{NT}$	Z_1^{NT}	$C_1^{M,NT}$	$C_1^{I,NT}$	$\bar{G}_2^{NT}$	Z_2^{NT}	$C_2^{M,NT}$	$C_2^{I,NT}$
(MVA)	(Ω)	(\$)	(\$)	(MVA)	(Ω)	(\$)	(\$)
12	0.16	2000	750000	15	0.13	3000	950000

and non-existing branches are provided in Table VI. Note that two conductor alternatives are available per branch. As in [9],

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TABLE VI
24-NODE SYSTEM—DATA FOR CANDIDATE CONDUCTORS

Type	Alternative 1			Alternative 2		
l	$\bar{F}_1^l$	Z_1^l	$C_1^{I,l}$	$\bar{F}_2^l$	Z_2^l	$C_2^{I,l}$
	(MVA)	(Ω/km)	(\$/\text{km})	(MVA)	(Ω/km)	(\$/\text{km})
<i>NRF</i>	6.28	0.557	19140	9.00	0.478	29870
<i>NAF</i>	3.94	0.732	15020	6.28	0.557	25030

TABLE VII
24-NODE SYSTEM—YEARLY LEVELS OF PRODUCTION AND LOSSES (GWh)

			Stage		
			1	2	3
Without	Production	Generators	0.00	0.00	0.00
		Transformers	107.54	178.24	302.78
DG	Losses	Feeders	1.68	2.29	4.14
		Transformers	0.31	0.45	0.87
With	Production	Generators	26.88	42.53	75.69
		Transformers	80.66	135.71	227.08
DG	Losses	Feeders	0.71	1.80	2.42
		Transformers	0.21	0.46	0.83

maintenance costs and lifetimes for all feeders are equal to \$450/year and 25 years, respectively.

The yearly investment budget is equal to \$6 million. Moreover, when investment in DG is allowed, the penetration limit, ξ , is set equal to 25%, which is well beyond the targeted level in Europe for 2020. In addition, the set of candidate nodes for DG installation are $\Omega^C = \{2, 3, 7, 13, 15, 16, 17, 20\}$, and $\Omega^W = \{1, 4, 5, 9, 15, 17, 18, 19\}$.

Under the aforementioned stopping criterion, CPLEX required 20.39 s for the case without DG and 12.01 s for the case with DG. This computational result reveals the case-dependent behavior of the branch-and-cut algorithm. The solutions for both cases are depicted in Fig. 2. As can be observed, the topologies of both distribution systems significantly differ at all stages. As another salient difference, capacities of feeders and transformers installed for the case with no DG are greater than or equal to those of the feeders installed for the case with DG, thereby incurring larger investment costs in feeders and transformers. In the case with DG, it is worth mentioning that the progressive increase in demand is partially compensated for by the installation of wind power generators, mainly at nodes belonging to zone A, where wind is more effective. Moreover, conventional generators are installed at load nodes outside zone A.

Table VII lists the yearly levels of production and energy losses for both cases. As can be noted, the incorporation of DG leads to a substantial reduction in the level of energy supplied by the transformers located at substation nodes. Moreover, for the case with DG, energy losses in feeders and transformers are lower than those with no DG, except for energy losses in transformers at stage 2, which are slightly higher for the case with DG even though the energy supplied by transformers is lower. This result stems from the larger impedance magnitudes of the transformers used in the case with DG.

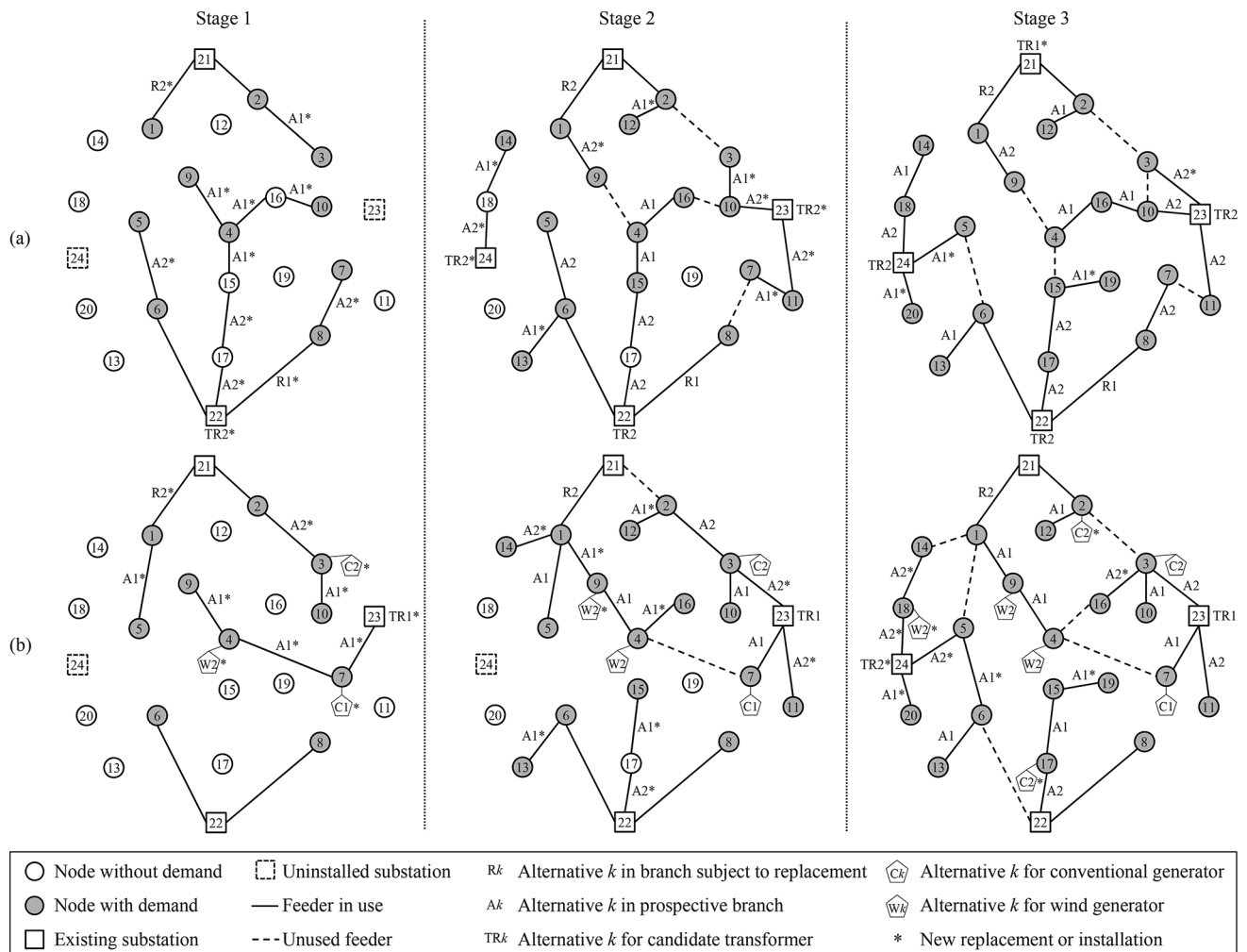


Fig. 2. Solutions for the 24-node system. (a) Without DG. (b) With DG.

TABLE VIII
24-NODE SYSTEM—YEARLY COSTS (10^6 \$)

		Stage		
		1	2	3
Without DG	Investment	1.36	2.47	0.94
	Maintenance	0.01	0.02	0.02
	Production	7.50	12.43	21.11
	Losses	0.14	0.19	0.35
	Unserved energy	0.00	0.00	0.00
With DG	Investment	5.82	4.51	5.63
	Maintenance	0.24	0.42	0.68
	Production	6.41	10.54	17.95
	Losses	0.07	0.16	0.23
	Unserved energy	0.00	0.00	0.00

Table VIII presents cost information on a yearly basis for both cases. As can be noted, for the case with no DG, the largest investment cost is incurred at stage 2 and corresponds to the construction of two new substations at nodes 23 and 24, including the installation of two new transformers at those nodes. For the case with DG, the largest investment is made at stage 1, in which 1) a substation and a transformer are added at candidate node 23, and 2) three DG units are installed at nodes 3, 4, and 7. In addition, for both cases, the operating costs related to maintenance, production, and losses rise along stages due to the in-

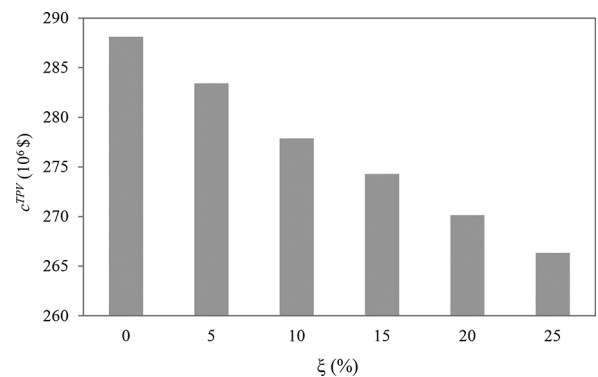


Fig. 3. Present value of the total cost versus penetration level of DG for the 24-node system.

crease in total demand. As can also be seen, no load shedding is required by both solutions. This result corroborates the suitability of the choice for C^U . Moreover, the incorporation of DG yields higher investment and maintenance costs. However, this expected cost increase is offset by the decrease in costs related to production and energy losses. Table IX lists the present values of the different costs associated with the solutions for both cases. It should be noted that a significant 7.55% reduction in the total cost is attained by the solution with DG.

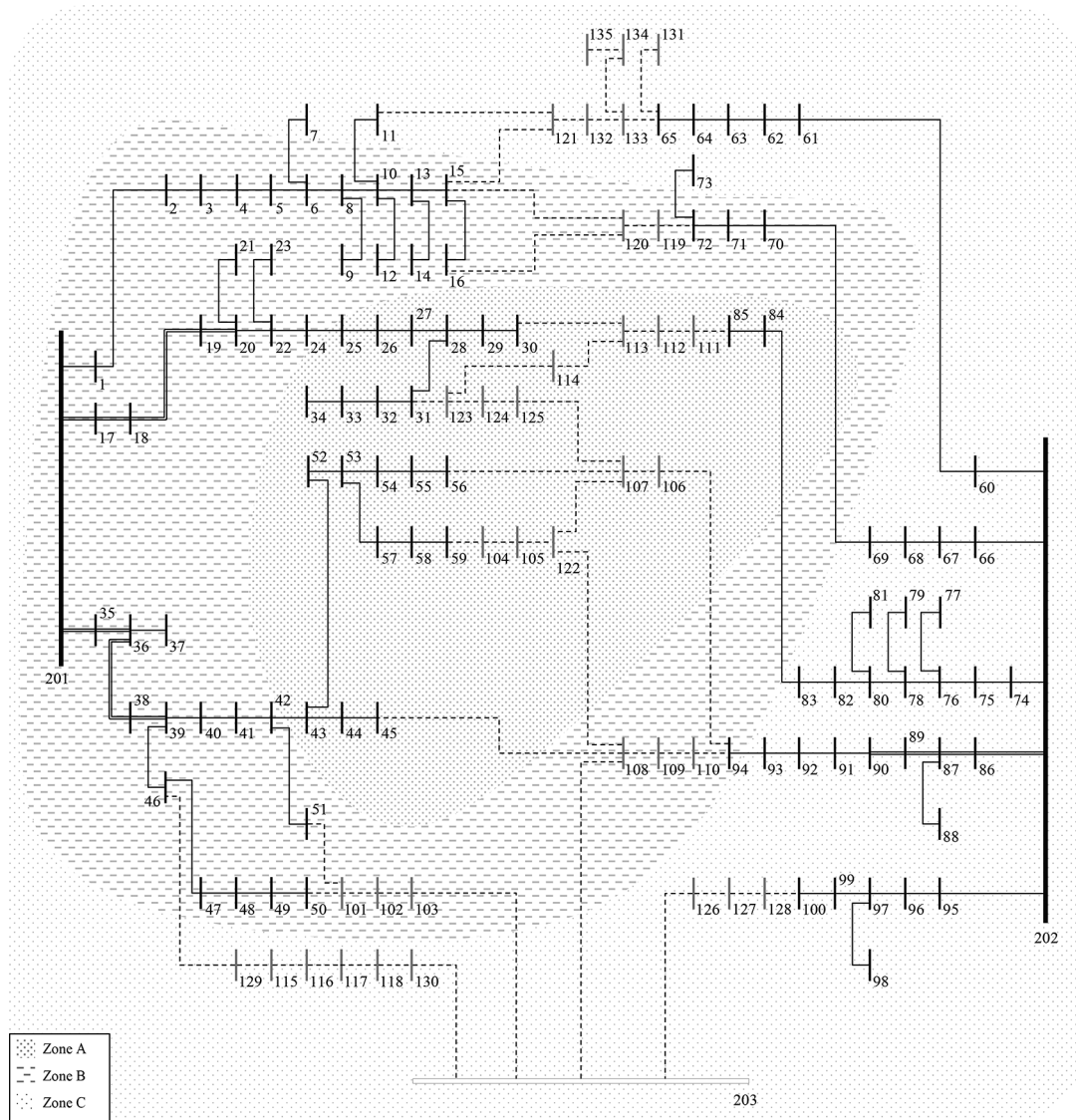


Fig. 4. One-line diagram of the 138-node system.

TABLE IX
24-NODE SYSTEM—PRESENT VALUES OF TOTAL COSTS (10^6 \$)

Case	Investment	Maintenance	Production	Losses	Unservd energy	c^{TPV}
Without DG	6.07	0.30	277.09	4.64	0.00	288.10
With DG	18.85	8.94	235.54	3.01	0.00	266.34

The model is also useful to assess the economic impact of the penetration level for DG. Fig. 3 shows the evolution of the present value of the total cost, c^{TPV} , for penetration levels ranging between 0% and 25%. Note that c^{TPV} decreases as the level of penetration increases.

B. 138-Node Case Study

The proposed MILP-based model has been applied to a 13.8-kV test system based on that described in [8] and depicted in Fig. 4. The system comprises 135 load nodes, represented

by short bars; 3 substation nodes, depicted as long bars; and 151 branches, indicated by lines. For reproducibility purposes, data for nodal peak demands and branch lengths can be downloaded from [33]. Voltages at substation nodes are set to 1.05 p.u. on a 13.8-kV base, whereas upper and lower bounds for voltages at load nodes are equal to 1.05 p.u. and 0.95 p.u., respectively. Candidate locations for distributed generation are $\Omega^C = \{10, 28, 38, 53, 64, 94, 108, 117, 126, 133\}$, and $\Omega^W = \{31, 52, 78, 94, 103, 113, 114, 116, 120, 122\}$.

Existing substations, which are located at nodes 201 and 202, include a 12-MVA transformer characterized by an impedance equal to 0.16Ω and a maintenance cost equal to \$2000. Node 203 represents the location of a candidate substation. Expansion costs of substations at nodes 201–203 are \$100 000, \$100 000, and \$150 000, respectively. Data for candidate transformers are provided in Table X, where the same two alternatives are available for each substation. The lifetime of all candidate transformers is 15 years.

The capacity and unitary impedance of existing feeders are 6.28 MVA and $0.557 \Omega/\text{km}$, respectively. Data for candidate

TABLE X
138-NODE SYSTEM—DATA FOR CANDIDATE TRANSFORMERS

Alternative 1				Alternative 2			
$\bar{G}_1^{NT}$	Z_1^{NT}	$C_1^{M,NT}$	$C_1^{I,NT}$	$\bar{G}_2^{NT}$	Z_2^{NT}	$C_2^{M,NT}$	$C_2^{I,NT}$
(MVA)	(Ω)	(\$)	(\$)	(MVA)	(Ω)	(\$)	(\$)
7.5	0.25	1000	500000	15	0.13	3000	950000

TABLE XI
138-NODE SYSTEM—DATA FOR CANDIDATE CONDUCTORS

Type	Alternative 1			Alternative 2		
	$\bar{F}_1^l$	Z_1^l	$C_1^{I,l}$	$\bar{F}_2^l$	Z_2^l	$C_2^{I,l}$
l	(MVA)	(Ω/km)	(\$/km)	(MVA)	(Ω/km)	(\$/km)
<i>NRF</i>	9.00	0.478	29870	12.00	0.423	39310
<i>NAF</i>	6.28	0.557	25030	9.00	0.478	34920

conductors in branches subject to replacement and non-existing branches are provided in Table XI. Note that two conductor alternatives are available per branch. As in [9], maintenance costs and lifetimes for all feeders are equal to \$450/year and 25 years, respectively.

For a yearly investment budget equal to \$5 million, CPLEX required 334.01 s for the case without DG and 907.86 s for the case with a 25% penetration limit for DG. Investment plans for both cases are reported in Table XII, where single figures denote nodes, pairs of figures connected by a hyphen denote feeders, and the symbols in brackets are consistent with the notation described in the legend of Fig. 2. Table XIII presents cost information on a yearly basis for both cases. As can be observed, for the case with no DG, the largest investments are made at stages 3 and 4 and are respectively associated with the expansion of the existing substation at node 202, and with the construction of a new substation at node 203. For the case with DG, the largest investments are made at stages 1, 5, and 9 and correspond to expansion decisions related to transformers and DG units (Table XII). In addition, the results for both cases are characterized by two relevant aspects also observed in the illustrative example: 1) unserved energy costs are zero, thereby validating the adequacy of C^U ; and 2) the operating costs related to maintenance, production, and losses rise along the time span as the total demand increases. Furthermore, when DG is allowed, the expected growth in investment and maintenance costs is compensated for by the reduction in production and energy losses costs. Table XIV lists the present values of the different costs for both cases. As can be seen, the total cost attained by the solution with DG represents a significant 7.56% reduction over that incurred without DG.

This case study is useful to substantiate the use of the proposed piecewise linear approximation for energy losses, which is motivated by the inability of CPLEX to find high-quality solutions with reasonable computational effort when quadratic terms are considered in the objective function. For the case with DG, CPLEX was unable to reduce the optimality gap under 50.7% after 50 000 s when the proposed model was implemented with quadratic losses. Moreover, the quality of the piecewise linearization has been assessed by solving the

TABLE XII
138-NODE SYSTEM—INVESTMENT PLANS

Stage	Without DG		With DG	
1	35-36(R1) 72-119(A1)	59-104(A1) 105-122(A1)	17-18(R2) 50-101(A1) 101-102(A1) 108-109(A1) 109-110(A1) 10(C2)	45-108(A1) 94-100(A1) 102-103(A1) 109-110(A1) 52(W2)
2	35-36(R1) 105-107(A1)	59-104(A1) 105-122(A1)	56-107(A1) 104-105(A1) 107-122(A1) 122(W2)	94-106(A1) 105-122(A1) 108-122(A2)
3	45-108(A2) 108-109(A1) 202(TR2)	51-101(A1) 109-110(A1)	31(W2)	
4	30-113(A1) 103-203(A2) 108-122(A1) 111-112(A1) 203(TR1)	85-115(A1) 104-105(A1) 108-203(A2) 113-114(A1)	31-123(A1) 112-113(A1) 114-123(A1)	84-111(A1) 113-114(A1)
5	46-129(A1) 115-129(A1) 117-118(A1) 130-203(A1)	115-116(A1) 116-117(A1) 118-130(A1)	103-203(A1) 108-203(A1) 115-116(A1) 117-118(A1) 123-124(A1) 130-203(A1) 203(TR1)	107-125(A2) 111-112(A1) 116-117(A1) 118-130(A1) 124-125(A1) 114(W2)
6	15-121(A2) 121-132(A1)	65-133(A1) 132-133(A1)	11-121(A1) 72-119(A1) 30-113(A1)	15-120(A1) 119-120(A1) 51-101(A1)
7	31-123(A1) 124-125(A1)	123-124(A1)	46-129(A1)	115-129(A1)
8	17-18(R1) 11-121(A1) 126-202(A1)	36-38(R1) 100-128(A1) 127-128(A1)	35-201(R1) 106-107(A2) 127-128(A1)	100-128(A1) 126-203(A1)
9	19-20(R1)	65-131(A1)	65-131(A1) 202(TR1)	113(W2)
10	17-18(R1) 86-87(R2) 87-89(R2) 134-135(A1)	17-201(R1) 86-202(R2) 133-134(A1)	18-19(R1) 132-133(A1) 134-137(A1)	65-133(A1) 133-134(A1)

problem with quadratic losses wherein binary investment and utilization variables were fixed to the values attained by the proposed mixed-integer linear program with a three-piece linearization for losses. For this particular case study, the total cost only differs by 0.35%. Such result validates the appropriateness of the piecewise linear approximation in terms of solution quality.

IV. CONCLUSION

This paper has investigated the incorporation of investment decisions on distributed generation in the multistage distribution expansion planning problem. The proposed tool allows the distribution planner to devise the least-cost investment strategy to address the estimated demand growth. As a significant modeling feature, the proposed problem formulation prevents the issues with isolated nodes and transfer nodes characterizing traditional radiality constraints when network expansion and distributed generation installation are co-optimized. The resulting model is formulated as a mixed-integer nonlinear programming problem. The subsequent use of some well-known linearization

TABLE XIII
138-NODE SYSTEM—YEARLY COSTS (10^6 \$)

		Stage									
		1	2	3	4	5	6	7	8	9	10
Without DG	Investment	0.14	0.05	1.11	0.83	0.16	0.06	0.03	0.09	0.03	0.08
	Maintenance	0.05	0.05	0.06	0.06	0.06	0.06	0.07	0.07	0.07	0.07
	Production	9.34	10.74	11.41	12.13	13.03	13.75	14.48	15.17	16.00	17.18
	Losses	0.21	0.29	0.30	0.25	0.38	0.31	0.37	0.37	0.41	0.46
	Unserved energy	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
With DG	Investment	4.41	3.47	3.39	0.09	4.25	0.16	0.05	0.06	4.01	0.06
	Maintenance	0.27	0.44	0.61	0.61	0.78	0.78	0.79	0.79	0.96	0.96
	Production	8.48	9.12	9.25	9.82	10.21	10.84	11.47	12.11	12.47	13.52
	Losses	0.16	0.18	0.18	0.21	0.22	0.21	0.22	0.26	0.26	0.29
	Unserved energy	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

TABLE XIV
138-NODE SYSTEM—PRESENT VALUES OF TOTAL COSTS (10^6 \$)

Case	Investment	Maintenance	Production	Losses	Unserved energy	c^{TPV}
Without DG	2.78	0.92	211.86	5.51	0.00	221.07
With DG	20.58	11.41	168.84	3.52	0.00	204.35

schemes yields a mixed-integer linear program suitable for efficient off-the-shelf software.

The proposed methodology has been successfully validated with two test systems. Numerical results reveal that incorporating distributed generation investment decisions in distribution planning leads to a substantial reduction in the total cost over the case with no generation investments. Moreover, simulations show that the computational performance of the proposed approach, albeit case dependent, is acceptable bearing in mind that a multistage planning problem is solved.

Ongoing research is focused on the stochastic nature of demand and wind. Further work will be devoted to the consideration of other generation technologies. Another interesting avenue of research is the analysis of the issues associated with distributed generation such as demand supply unbalance in a fault condition, power quality decline, and reduction of reliability levels.

APPENDIX

Based on [34], a piecewise linear approximation is used for the quadratic terms in (5) representing energy losses in transformers and feeders. Thus, expressions (5) are replaced by

$$c_t^R = \sum_{b \in B} \Delta_b C_b^{SS} pf \left[\sum_{tr \in TR} \sum_{k \in K^{tr}} \sum_{s \in \Omega^{SS}} \sum_{\nu=1}^{n_\nu} M_{k\nu}^{tr} \delta_{sktb\nu}^{tr} + \sum_{l \in L} \sum_{k \in K^l} \sum_{(s,r) \in T^l} \sum_{\nu=1}^{n_\nu} M_{k\nu}^l \ell_{sr} (\delta_{srktb\nu}^l + \delta_{rsktb\nu}^l) \right]; \quad \forall t \in T \quad (38)$$

$$g_{sktb}^{tr} = \sum_{\nu=1}^{n_\nu} \delta_{sktb\nu}^{tr}; \quad \forall tr \in TR, \forall s \in \Omega^{SS}, \forall k \in K^{tr}, \forall t \in T, \forall b \in B \quad (39)$$

$$0 \leq \delta_{sktb\nu}^{tr} \leq A_{k\nu}^{tr}; \quad \forall \nu = 1 \dots n_\nu, \quad \forall tr \in TR, \forall s \in \Omega^{SS}, \forall k \in K^{tr}, \forall t \in T, \forall b \in B \quad (40)$$

$$f_{srktb}^l = \sum_{\nu=1}^{n_\nu} \delta_{srktb\nu}^l; \quad \forall l \in L, \forall s \in \Omega_r^l, \forall r \in \Omega^N, \forall k \in K^l, \forall t \in T, \forall b \in B \quad (41)$$

$$0 \leq \delta_{srktb\nu}^l \leq A_{k\nu}^l; \quad \forall \nu = 1 \dots n_\nu, \quad \forall l \in L, \forall s \in \Omega_r^l, \forall r \in \Omega^N, \forall k \in K^l, \forall t \in T, \forall b \in B \quad (42)$$

where (38) are the linearized costs of energy losses, while (39)–(40) and (41)–(42) are related to the linearization of energy losses in transformers and feeders, respectively.

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